

BrineX - Commercialization Plan

1. Target Market

Primary Users:

- 10,000+ salt pan landowners (Puttalam, Hambantota)
- 20+ SLSI-regulated quality testing laboratories
- 100+ registered salt distributors
- Regional salt societies (PSS - Puttalam Salt Society)

Market Size:

- Global salt market valued at \$28.9B (2024)
- Asia-Pacific contributes 35% of global market
- Sri Lanka employs 30,000+ workers across the salt production value chain
- No existing digital platform targets this vertical

Expansion Path:

- Phase 1: Puttalam region (largest salt production hub in Sri Lanka)
- Phase 2: Hambantota and all major Sri Lankan salt-producing regions
- Phase 3: South Asian tropical markets
- Phase 4: Global

2. Value Proposition

"Replacing guesswork with AI - from harvest to quality to trade."

Why better than alternatives:

Current Alternative	Limitation	BrineX Advantage
Manual quality inspection	Subjective, inconsistent, no records	YOLOv8 real-time 3-class detection - objective, instant, auditable
Experience-based production planning	No data, reactive decisions	LSTM parameter prediction + Linear Regression 2-month production forecasts
Phone-based buyer-seller negotiation	No price transparency, no deal tracking	Digital marketplace - sellers ranked by highest price and bag count, SARIMAX price prediction
Paper logbooks	Lost records, no analytics	Cloud-stored measurements, accessible from any device
Separate tools for each function	Data silos, multiple logins	Single integrated platform covering all production stages
Generic farm management software	Not designed for salt crystallization	Domain-specific: 8 crystallization parameters, salt-specific ML models

3. Revenue Model

Tiered SaaS Subscription:

Free

Rs 0 /forever

For individual landowners getting started with smart salt production.

✓ Basic crystallization alerts

✓ 7-day weather integration

✓ Single saltern monitoring

✓ Community support

✓ Basic production logs

Get Started Free

For Landowner

MOST POPULAR

Pro

Rs 2999 /per month

Full Crystal + Compass access for commercial salt producers scaling operations.

✓ Advanced crystallization prediction

✓ 30-day forecasting window

✓ Unlimited salterns

✓ Season planning dashboards

✓ Production analytics

✓ Priority email support

✓ Export data & reports

Start Pro Trial

For Landowner, Distributor

Lab

Rs 4999 /per month

Everything in Pro plus Vision AI for laboratories and quality-focused operations.

✓ Vision AI salt detection

✓ Purity classification reports

✓ Live camera integration

✓ Batch quality management

✓ Dedicated support

✓ API access

Start Lab Plan

For Laboratory

Payment Infrastructure:

- PayHere payment gateway - secure LKR payments
- Automatic subscription lifecycle management (active, expired, downgrade)
- Plan-level feature gating enforced at API level

4. Cost and Feasibility

Development Cost: Completed - full-stack platform built and functional with 9 backend microservices, modern frontend, and 3 trained ML models (LSTM, SARIMAX, YOLOv8).

Monthly Operational Cost (On-Demand Production):

Item	Monthly Cost (LKR)	Notes
AWS EC2 (ECS - 2× t3.xlarge)	~33,600	Runs all 11 containerized services
AWS EC2 (Kafka - 1× t3.medium)	~4,200	Self-hosted Kafka + Zookeeper
AWS ALB	~3,500	Load balancer + WebSocket support
AWS NAT Gateway	~4,900	Required for private subnet outbound
AWS misc (ECR, SQS, CW, Route53, Secrets)	~1,500	Low-cost managed services
MongoDB Atlas (M10)	~8,000	Shared production cluster

SMS OTP (Notify.lk)	~3,000	~1,000 OTPs/month
Domain + SSL	~500	.lk domain + free ACM cert
Total	~59,200	

Cost Optimization Strategies:

Optimization	Savings
Reserved Instances (1-year commit)	30–60% on EC2 → saves ~LKR 13,000–22,000
Spot Instances for non-critical services (email, audit-log)	Up to 70% on those tasks
Downsize to t3.large if load permits early on	~\$120/month instead of \$240 → saves ~LKR 16,800
Reduce to 1 ECS instance at launch	Halves EC2 cost → saves ~LKR 16,800

Optimized monthly cost: ~LKR 37,000–42,000

5. Breakeven Analysis

On-Demand Breakeven (~LKR 59,200/month):

Scenario	Pro Subscribers	Lab Subscribers	Revenue (LKR)	Covers Cost?
Pro only	20	0	59,980	Yes
Lab only	0	12	59,988	Yes
Balanced	10	6	59,984	Yes
Conservative	8	5	48,987	No (needs optimization)
Realistic early	12	5	60,983	Yes

With Cost Optimization (~LKR 37,000–42,000/month):

Scenario	Pro Subscribers	Lab Subscribers	Revenue (LKR)	Covers Cost?
Minimal	7	4	40,989	Yes
Conservative	8	3	38,989	Yes
Safe buffer	10	5	54,985	Yes (with profit)

6. Revenue Projections

Year	Free Users	Pro Subscribers	Lab Subscribers	Monthly Revenue (LKR)
Year 1	500	50	10	~100,000
Year 2	2,000	200	40	~400,000
Year 3	5,000	500	100	~1,000,000

7. Competitive Advantages

- First-mover advantage - no existing digital platform targets the salt production vertical
- Full value chain coverage - production, quality, marketplace, and analytics in one platform
- AI at the core - LSTM parameter prediction, Linear Regression production forecasting, YOLOv8 quality detection, SARIMAX price prediction, and Federated Learning waste valorization
- Seller transparency - recommendations driven by highest price and bag count, not hidden algorithms
- Freemium onboarding - low barrier to entry with 14-day Pro trial
- Scalable infrastructure - microservices and Docker handle 100 to 100,000 users
- Compliance-ready - audit logging, KYC onboarding, role-based access control
- Domain-specific - built exclusively for salt crystallization, not adapted from generic tools